

A candidate counterexample to Conjecture 2 in Jones’ Conjecture in subcubic graphs

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A 12-vertex planar graph has $\text{fvs} = 5$ and $\text{fp} = 2$; a related family has fvs/fp tending to 3. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 2](#) from *Jones’ Conjecture in subcubic graphs*.

1.1 Problem statement

For every planar graph G , we have $\text{fvs}(G) \leq 2 \cdot \text{fp}(G)$, where $\text{fp}(G)$ is the maximum size of a face-packing of G , i.e., a cycle-packing where, for some embedding of G , every cycle bounds a face.

1.2 Catalog context

The authors pose this stronger conjecture to highlight how little is understood around Jones’ Conjecture. They note that the example of many nested disjoint cycles shows the embedding cannot be fixed, and that the simple discharging argument giving $\text{fvs}(G) \leq 3 \text{cp}(G)$ does not imply this conjecture even with factor 3 instead of 2.

1.3 Claimed result

Model verdict: disproved (confidence: high).

A 12-vertex planar graph has $\text{fvs} = 5$ and $\text{fp} = 2$; a related family has fvs/fp tending to 3.

The model’s own caveat: Uses the stated vertex-disjoint, planar-embedding definition; no claim of minimality or literature priority.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 Counterexample theorem

The conjecture is false as stated. More strongly, for every integer $t \geq 1$, there is a simple planar graph G_t , of maximum degree four and order $8t + 4$, such that

$$\text{fvs}(G_t) = 3t + 2 \quad \text{and} \quad \text{fp}(G_t) = t + 1.$$

Consequently,

$$\frac{\text{fvs}(G_t)}{\text{fp}(G_t)} = 3 - \frac{1}{t + 1}.$$

For $t = 1$, this gives a 12-vertex counterexample:

$$\text{fvs}(G_1) = 5 > 4 = 2 \text{fp}(G_1).$$

The important point is that the upper bound on face-packing below holds **in every planar embedding**, not merely in a chosen drawing.

2.2 1. Construction

Take pairwise vertex-disjoint cycles

$$C_i = a_i b_i c_i d_i a_i, \quad 1 \leq i \leq t,$$

and pairwise vertex-disjoint copies

$$Q_j = G[\{p_j, q_j, r_j, s_j\}] \cong K_4, \quad 0 \leq j \leq t.$$

All these vertex sets are mutually disjoint.

Add exactly the following edges between them:

- $a_i p_i$ and $c_i q_i$, for $1 \leq i \leq t$;
- $d_i b_{i+1}$, for $1 \leq i < t$;
- $b_1 p_0$ and $d_t q_0$.

Thus the C_i 's form a chain, each C_i has a K_4 attached at the opposite vertices a_i, c_i , and Q_0 connects the two ends of the chain.

There are

$$4t + 4(t + 1) = 8t + 4$$

vertices. Every vertex has degree three except the vertices p_j, q_j , which have degree four.

For $t = 1$, the construction is particularly simple: take a square $abcd$ and two disjoint K_4 's. Connect one K_4 to a, c , and the other to b, d , using distinct attachment vertices in each K_4 .

2.2.1 Planarity

First consider the auxiliary graph consisting of:

- the cycles C_i ;
- the chain edges $d_i b_{i+1}$;
- a closing edge $d_t b_1$;
- the diagonals $a_i c_i$.

Draw the C_i 's as diamonds in a row, with b_i, d_i their left and right corners. Draw each diagonal $a_i c_i$ inside its diamond and the closing edge around the outside of the entire chain. This is a planar drawing.

Now replace each diagonal $a_i c_i$ by the two-terminal gadget Q_i , with connecting edges $a_i p_i, c_i q_i$, and replace the closing edge by Q_0 , with connecting edges $b_1 p_0, d_t q_0$.

These replacements preserve planarity: the two terminals p_j, q_j lie on a common face of a plane K_4 , so each replacement can be performed in a small neighborhood of the corresponding edge.

Choose the embedding of each Q_j with outer triangle $p_j q_j r_j$ and s_j inside. Its three internal triangular faces remain faces after the replacement. In particular,

$$p_j q_j s_j p_j, \quad 0 \leq j \leq t,$$

are pairwise vertex-disjoint facial cycles in this embedding. Therefore

$$\text{fp}(G_t) \geq t + 1. \quad (1)$$

2.3 2. Exact feedback vertex number

Every feedback vertex set must contain:

- at least one vertex of each C_i ;
- at least two vertices of each $Q_j \cong K_4$.

The latter assertion follows because any three vertices of a K_4 induce a triangle. These subgraphs have mutually disjoint vertex sets, so

$$\text{fvs}(G_t) \geq t + 2(t + 1) = 3t + 2. \quad (2)$$

For the reverse inequality, delete

$$S = \{a_i : 1 \leq i \leq t\} \cup \{p_j, q_j : 0 \leq j \leq t\}.$$

The remaining vertices of the C_i 's form the single path

$$b_1 c_1 d_1 b_2 c_2 d_2 \cdots b_t c_t d_t,$$

and each Q_j leaves only the isolated edge $r_j s_j$. All gadget-attachment edges have been removed. Thus $G_t - S$ is a forest, proving

$$\boxed{\text{fvs}(G_t) = 3t + 2.} \quad (3)$$

2.4 3. A K_4 gadget can be met by only one packed cycle

Fix j . The only edges between Q_j and the rest of G_t are its two attachment edges, incident with p_j and q_j .

Consider any simple cycle meeting Q_j .

- If it is contained in Q_j , it uses at least three of its four vertices.
- Otherwise, it must use both attachment edges, and hence contains both p_j and q_j .

It follows that **any two cycles meeting Q_j intersect**:

- two cycles contained in Q_j intersect because each uses at least three vertices;
- two cycles leaving Q_j both contain p_j, q_j ;
- a cycle contained in Q_j intersects any cycle containing p_j, q_j , since a three-element subset of a four-element set meets every two-element subset.

Therefore, in any vertex-disjoint cycle-packing, at most one cycle can meet a given Q_j . This observation is independent of the embedding.

2.5 4. None of the cycles C_i can ever be facial

Fix i . There is an a_i - c_i path

$$P_i = a_i p_i q_i c_i$$

whose internal vertices lie in Q_i .

There is also a b_i - d_i path R_i whose internal vertices avoid both C_i and Q_i : follow the chain from b_i towards b_1 , cross through

$$b_1 p_0 q_0 d_t,$$

and follow the other end of the chain back to d_i . When traversing another C_j , use its path $b_j c_j d_j$.

The paths P_i and R_i are internally vertex-disjoint, and their endpoints alternate around

$$C_i = a_i b_i c_i d_i a_i.$$

In any planar embedding, C_i is a Jordan curve. Two internally disjoint paths with alternating endpoints on this curve cannot both lie on the same side: a path joining a_i to c_i separates that side into regions incident with b_i and d_i , respectively.

Hence P_i and R_i lie on opposite sides of C_i in **every** planar embedding. Both sides therefore contain graph edges, so

$$\boxed{C_i \text{ is not facial in any planar embedding of } G_t.} \quad (4)$$

Equivalently, $C_i \cup P_i \cup R_i$ is a subdivision of K_4 , with C_i corresponding to a four-cycle rather than a facial triangle.

2.6 5. Exact face-packing number

Delete the vertices of all gadgets Q_j . What remains is the chain of cycles $C_1, \dots, C_t$, joined by the edges $d_i b_{i+1}$. Those joining edges are bridges in the remaining graph. Consequently, its only simple cycles are the C_i 's.

By (4), none of those cycles can be facial in any planar embedding of the full graph. Thus **every facial cycle of G_t must meet at least one gadget Q_j** .

By Section 3, different members of a vertex-disjoint face-packing cannot meet the same gadget. There are only $t + 1$ gadgets, so every planar embedding satisfies

$$\text{maximum face-packing size} \leq t + 1.$$

Together with the explicit facial packing in (1), this proves

$$\boxed{\text{fp}(G_t) = t + 1.} \quad (5)$$

2.7 6. Consequences and scope

Taking $t = 1$ proves the claimed disproof:

$$\text{fvs}(G_1) = 5, \quad \text{fp}(G_1) = 2.$$

Moreover, the family shows that **no universal inequality**

$$\text{fvs}(G) \leq c \text{fp}(G)$$

can hold for all planar graphs with any constant $c < 3$. This does not establish an upper bound with constant three; it only shows that such a constant would be best possible if an upper bound exists.

For comparison,

$$\text{cp}(G_t) = 2t + 1.$$

The lower bound comes from taking all C_i 's and one triangle from each Q_j . The upper bound follows from Section 3 and the fact that the only cycles avoiding all gadgets are the C_i 's. Thus these examples do **not** contradict the original cycle-packing version of Jones' conjecture.

The argument is self-contained and handles all planar embeddings. I have not checked whether this construction has previously appeared in the literature.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	1912.01570__00
Catalog entry	Conjecture 2
Source corpus	arxiv
Campaign leg	previously unattacked arXiv records (attacks_arxiv_astra)
Source paper / entry	Jones' Conjecture in subcubic graphs
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_arxiv_astra/1912.01570__00/output.md
Independent review	None
arXiv	arXiv:1912.01570
Model's caveats	Uses the stated vertex-disjoint, planar-embedding definition; no claim of minimality or literature priority.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/1912.01570__00/`.